

Sampling and Bootstrapping, MLE, Beta

Before you leave lab, make sure you [click here](#) so that you're marked as having attended this week's section. The CA leading your discussion section can enter the password needed once you've submitted.

1 Warmups

1.1 MLE

Suppose $x_1, \dots, x_n$ are iid (independent and identically distributed) values sampled from some distribution with density function $f(x|\theta)$, where θ is unknown. Recall that the likelihood of the data is

$$L(\theta) = f(x_1, x_2, \dots, x_n | \theta) = \prod_{i=1}^n f(x_i | \theta)$$

Recall we solve an optimization problem to find $\hat{\theta}$ which maximizes $L(\theta)$, i.e., $\hat{\theta} = \arg \max_{\theta} L(\theta)$.

- Write an expression for the log-likelihood, $LL(\theta) = \log L(\theta)$.
- Why *can* we optimize $LL(\theta)$ rather than $L(\theta)$?
- Why *might* we optimize $LL(\theta)$ rather than $L(\theta)$?

- $\sum_{i=1}^n \log f(x_i | \theta)$
- Logarithms are strictly increasing functions. For any strictly increasing function f and any function g , the following holds: $\arg \max g(x) = \arg \max f(g(x))$
- Finding the max of a function requires you take derivatives. The original expression consists of a product of many functions of θ . This leads to a difficult derivation because of the chain rule of calculus.

By taking the log, we convert the product into sums, and the derivative of the sum of many functions of θ is much easier to compute.

That is, $\frac{\partial}{\partial \theta} \sum_i f_i(\theta) = \sum_i \frac{\partial}{\partial \theta} f_i(\theta)$ which is easy.

However, $\frac{\partial}{\partial \theta} \prod_i f_i(\theta) =$ a complicated expression

1.2 Beta

- Suppose you have a coin where you have no prior belief on its true probability of heads p . How can you model this belief as a Beta distribution?
- Suppose you have a coin which you believe is fair, with "strength" α . That is, pretend you've seen α heads and α tails. How can you model this belief as a Beta distribution?
- Now suppose you take the coin from the previous part and flip it 10 times. You see 8 heads and 2 tails. How can you model your posterior belief of the coin's probability of heads?

- Beta(1, 1) is a uniform prior, meaning that prior to seeing the experiment, all probabilities of heads are equally likely.
- Beta($\alpha + 1$, $\alpha + 1$). This is our prior belief about the distribution.
- Beta($\alpha + 9$, $\alpha + 3$)

1.3 Beta Sum

What is the distribution of the sum of 100 iid Betas? Let X be the sum

$$X = \sum_{i=1}^{100} X_i \quad \text{where each } X_i \sim \text{Beta}(a = 3, b = 4)$$

Note the variance of a Beta:

$$\text{Var}(X_i) = \frac{ab}{(a+b)^2(a+b+1)} \quad \text{where } X_i \sim \text{Beta}(a, b)$$

By the Central Limit Theorem, the sum of equally weighted IID random variables will be Normally distributed. We calculate the expectation and variance of X_i using the beta formulas:

$$\begin{aligned} E(X_i) &= \frac{a}{a+b} && \text{Expectation of a Beta} \\ &= \frac{3}{7} \approx 0.43 \end{aligned}$$

$$\begin{aligned} \text{Var}(X_i) &= \frac{ab}{(a+b)^2(a+b+1)} && \text{Variance of a Beta} \\ &= \frac{3 \cdot 4}{(3+4)^2(3+4+1)} \\ &= \frac{12}{49 \cdot 8} \approx 0.03 \end{aligned}$$

$$\begin{aligned} X &\sim N(\mu = n \cdot E[X_i], \sigma^2 = n \cdot \text{Var}(X_i)) \\ &\sim N(\mu = 43, \sigma^2 = 3) \end{aligned}$$

2 Problems

2.1 Variance of Hemoglobin Levels

A medical researcher treats patients with dangerously low hemoglobin levels. She has formulated two slightly different drugs and is now testing them on patients. First, she administered drug A to one group of 50 patients and drug B to a separate group of 50 patients. Then, she measured all the patients' hemoglobin levels post-treatment. For simplicity, assume that all variation in the patient outcomes is due to their different reactions to treatment.

The researcher notes that the sample mean is similar between the two groups: both have mean hemoglobin levels around 10g/dL. However, drug B's group has a **sample variance** that is 3 (g/dL)² **greater** than drug A's group. The researcher thinks that patients respond to drugs A and B differently. Specifically, she wants to make the scientific claim that drug A's patients will end up with a significantly different spread of hemoglobin levels compared to drug B's.

You are skeptical. It is possible that the two drugs have practically identical effects and that the observed different in variance was a result of chance and a small sample size, i.e. the **null hypothesis**. Calculate the probability of the null hypothesis using bootstrapping. Here is the data. Each number is the level of an independently sampled patient:

Hemoglobin Levels of Drug A's Group ($S^2 = 6.0$): 13, 12, 7, 16, 9, 11, 7, 10, 9, 8, 9, 7, 16, 7, 9, 8, 13, 10, 11, 9, 13, 13, 10, 10, 9, 7, 7, 6, 7, 8, 12, 13, 9, 6, 9, 11, 10, 8, 12, 10, 9, 10, 8, 14, 13, 13, 10, 11, 12, 9

Hemoglobin Levels of Drug B's Group ($S^2 = 9.1$): 8, 8, 16, 16, 9, 13, 14, 13, 10, 12, 10, 6, 14, 8, 13, 14, 7, 13, 7, 8, 4, 11, 7, 12, 8, 9, 12, 8, 11, 10, 12, 6, 10, 15, 11, 12, 3, 8, 11, 10, 10, 8, 12, 8, 11, 6, 7, 10, 8, 5

Complete the prompts in this [Colab notebook](#) to investigate this question using bootstrapping.

You can visit [this notebook](#) to see how we fleshed out all of the different functions expected of you for this problem.

2.2 Parameter Estimation and Wealth Distribution

The broader field of economics also relies on likelihood estimation and parameter estimation. One continuous probability distribution—one with a long tail as x approaches infinity—is used to model wealth inequality and the socioeconomic problems that stem from it. This probability distribution is given as:

$$f(x|\omega) = \frac{\omega 3^\omega}{x^{\omega+1}}, \text{ where } x \geq 3, \omega > 1$$

Assume that you’ve observed a sample of iid random variables $(X_1, X_2, X_3, \dots, X_n)$, where each of the X_i is modeled according to the above probability distribution function.

- a. What is the log-likelihood function $LL(\omega)$ of the sample $(X_1, X_2, X_3, \dots, X_n)$? Simplify using properties of logarithms wherever possible.

$$\begin{aligned} L(\omega) &= \prod_{i=1}^n \frac{\omega 3^\omega}{x_i^{\omega+1}} \\ LL(\omega) &= \sum_{i=1}^n \log \frac{\omega 3^\omega}{x_i^{\omega+1}} \\ &= \sum_{i=1}^n \log \omega + \sum_{i=1}^n \log 3^\omega - \sum_{i=1}^n \log x_i^{\omega+1} \\ &= \log \omega \sum_{i=1}^n 1 + \omega \sum_{i=1}^n \log 3 - \omega \sum_{i=1}^n \log x_i - \sum_{i=1}^n \log x_i \\ &= n \log \omega + n\omega \log 3 - \omega \sum_{i=1}^n \log x_i - \sum_{i=1}^n \log x_i \end{aligned}$$

- b. Set up the equation that would need to be solved in order to compute $\hat{\omega}_{MLE}$. Once you arrive at the equation and have worked through any calculus, you can stop and simply present the equation that can be solved via simple algebraic manipulation.

Computing $\hat{\omega}_{MLE}$ amounts to finding the value of ω that solves the equation $\frac{\delta LL(\omega)}{\delta \omega} = 0$. Fortunately, the derivative of interest is easily computed, because our $LL(\omega)$ is a sum of

logarithms and linear terms.

$$LL(\omega) = n \log \omega + n\omega \log 3 - \omega \sum_{i=1}^n \log x_i - \sum_{i=1}^n \log x_i$$

$$\frac{\delta LL(\omega)}{\delta \omega} = \frac{n}{\omega} + n \log 3 - \sum_{i=1}^n \log x_i = 0$$

You were welcome to stop at the above equation, since it can be solved via algebraic manipulation you needn't show. We do, however, present the solution below for completeness, since this is a real probability distribution used in many fields of statistical analysis.

$$\frac{n}{\hat{\omega}_{MLE}} = S_{\log X} - n \log 3, \text{ where } S_{\log X} = \sum_{i=1}^n \log x_i$$

$$\hat{\omega}_{MLE} = \frac{n}{S_{\log X} - n \log 3}$$

2.3 Dirk/Evan Showdown

Dirk and Evan are set to compete in a match of seven games, and the player winning the most will be rewarded with a medal from the Ruler of all of Bayestopia, Queen Doris. The outcomes of each of the seven games are independent of one another, and Dirk wins each game with a probability p (so that Evan wins with probability $1 - p$). Unfortunately, p is unknown, so you decide to model p itself as a Beta random variable such that $p \sim \text{Beta}(1, 1)$.

To learn more about p , you read up on all of their previous games, find that they've already competed 12 times, and learn that Evan has won 7 of those 12 games in this order: WLLWWLWWLWW. Even if Evan's edge is ever so slight, he appears to be the a priori favorite.

- Find the posterior distribution of p given Dirk and Evan's prior history competing against one another.
- Recall that the PDF of a *Beta* distribution on integer parameters a and b is given as:

$$f(x|a, b) = \frac{1}{B(a, b)} x^{a-1} (1-x)^{b-1}, \text{ where } B(a, b) \text{ is a normalization constant}$$

Assuming our hyperparameters a and b are positive integers, explain why the expected value of $p^m(1-p)^n$ —that is, $E[p^m(1-p)^n]$ —is given by

$$E[p^m(1-p)^n] = \frac{B(a+m, b+n)}{B(a, b)}$$

You can assume that m and n are nonnegative integers as well.

- c. Relying only on the posterior distribution of p that you computed for part a and the result from part b, compute the expected probability that the best-of-seven match between Dirk and Evan isn't settled after the first six games and therefore requires the seventh be played?

- a. Because the prior is $p \sim \text{Beta}(1, 1)$, and because the likelihood function is a Binomial with $n = 12$ and unknown p , our posterior is also a Beta, but with parameters of $a = 1 + 5 = 6$ and $b = 1 + 7 = 8$. More compactly, $p|\text{data} \sim \text{Beta}(6, 8)$.

- b. LOTUS tells us that the expected value of an arbitrary function $f(x)$ is:

$$E[f(x)] = \frac{1}{B(a, b)} \int_0^1 f(x) x^{a-1} (1-x)^{b-1}.$$

$f(x) = x^m(1-x)^n$ blends beautifully with the PDF of our Beta posterior:

$$\begin{aligned} E[f(p)] &= E[p^m(1-p)^n] = \frac{1}{B(a, b)} \int_0^1 x^m(1-x)^n x^{a-1} (1-x)^{b-1} \\ &= \frac{1}{B(a, b)} \int_0^1 x^{m+a-1} (1-x)^{n+b-1} = \frac{B(a+m, b+n)}{B(a, b)} \end{aligned}$$

- c. The probability that a seventh game is played is $\binom{6}{3} p^3(1-p)^3$. So the expected value is $\binom{6}{3} p^3(1-p)^3$ would be:

$$\binom{6}{3} \frac{B(9, 11)}{B(6, 8)}.$$